

Salient Characteristics

For

Tactical Data Networks Lab Tech Refresh Satellite Simulators

AFLCMC/C3NE, Hanscom AFB, MA 01731



Date: 29 April 2026

Prepared by:

AFLCMC/C3NE

Hanscom AFB, MA. 01731

Overview:

This Statement of Work (SOW) consists of the purchase of three types of satellite simulator/emulator hardware and accessories for the TDN Lab Refresh project. The equipment must meet the requirements stated in this SOW.

The first type of emulator needed is a multichannel UHF MUOS satellite simulator.

The second type of emulator needed is a multichannel UHF DAMA/IW satellite simulator.

The third type of emulator needed is a GPS satellite simulator.

Location:

Material shall be delivered to:

AFLCMC/C3NE
15 Eglin Rd Bldg. 1607
Lantern (TDN LAB)
Hanscom AFB, MA 01731
Attn: Greg Hiller, 781-225-6636

Points of Contact:

TDN Lab Manager

Primary contact:

Name: Louis Hegger
Email: louis.hegger@us.af.mil
Telephone: 781-225-4393

Secondary contact:

Name: Greg Hiller
Email: gregory.hiller.1.ctr@us.af.mil
Telephone: 781-225-6636

Scope

The purpose of this effort is to procure specialized tools and support equipment replacing aging support assets within the TDN Lab.

General Requirement:

- Vendor shall deliver all radio equipment and ancillary equipment on this order to the delivery location specified in this SOW. Any installation and assembly will be completed by TDN Lab personnel.
- Vendor shall ensure all owners' manuals, warranty information, maintenance and care instructions, and any other information/documentation is provided to the acceptor.

Note: Vendor shall be responsible for all damages incurred during shipping and discovered upon customer receipt/assembly.

Delivery Date:

Delivery shall be best effort. If there are any delivery issues, the vendor shall proactively address them with customers and contracting officer. The vendor shall provide notification to the government ordering representative at least 5 days prior to delivery.

Requirements:

MUOS Satellite Emulator:

The MUOS satellite emulator must be capable of the following functions:

- Portable and durable design
- Emulates MUOS system operation without using the live satellite
- Supports up to 4 simultaneous call service legs and up to 30 concurrent netted users
- Support user created custom scenarios
- Default scenarios included
- Simulate hard limited channels
- Supported Call Types:
 - Point-to-Point (PtP)
 - Point-to-Net (PtN)
 - Netted Comms (Group)
 - Multi-RAB
- Data Rates:
 - 2.4 kbps
 - 9.6 kbps
 - 32 kbps

- 64 kbps
- Operating Frequencies:
 - 300-320 MHz (UHF) in the U2B direction (configurable)
 - 360-380 MHz (UHF) in the B2U direction (configurable)
 - Hardware supports MUOS Plan B frequencies
- Control and Interface:
 - Web based Graphical User Interface
 - Local and Remote Access
 - Supported
- Power:
 - 120VAC
- GPS input
- Management software availability
- Equipment training is desired

DAMA/IW Satellite Emulator:

- Portable and durable design
- Emulates the IW or DAMA system operation without using the live satellite
- Operating modes
 - IW 5/25 kHz
 - DAMA/DASA 5 kHz, 25 kHz supported Dedicated Access and other waveforms
- Satellite simulation functions
 - Frequency translation
 - Digital band-limiting and hard-limiting
 - Variable digital propagation delay
 - Variable digital thermal noise
- Channel Simulation
 - 1 to 4 UHF SATCOM channels
- Encrypted order wire
- Operating frequencies:
 - Transmit Frequency 292 to 318 MHz
 - Receive Frequency 243 to 270 MHz
- User Interface (Direct Connection)
 - 1 to 4 user terminals in addition to local RF (direct connection) connected user terminals
- RF input power protection up to 250 W

- User Baseband I/O
- Serial I/O ports per channel with MIL-STD-188-114 and RS-232/RS-422 interface
- Interoperable with KY-57, KY-58, KY-99, KY-100, KIV-7, KYV-5, KG-84

- User I/O Rates (bps)
 - IW 75, 300, 600, 1200, 2400, 4800, 6000, 7200, 8K, 9.6K, 16K, 19.2K, 28.8K, 32K, 38.4K, 48K, 56K
 - 5 kHz DAMA 75, 300, 600, 1200, 2400
 - 25 kHz DAMA 75, 300, 600, 1200, 2400, 4800, 16K
 - Non-DAMA 1200, 2400, 4800, 6000, 7200, 8000, 9600, 19.2K, 16K, 28.8K, 32K, 38.4K, 48K, 56K
- MIL-STD Interfaces desired
 - MIL-STD-188-181B/C
 - MIL-STD -182A
 - MIL-STD -183B
 - MIL-STD -183
 - MIL-STD -183A
 - MIL-STD -185A
- Modulations supported: SOQPSK, BPSK, DEQPSK, (S)BPSK, FSK, CPM
- Cryptographic Keyfill: KYK-13, KYX-15, KOI-18, AN/CYZ-10, SKL
- External Reference 1, 5, or 10 MHz
- Management software availability
- Equipment training is desired

GPS Satellite Emulator:

- Key Simulation Features
 - 1000 Hz simulation iteration rate or greater
 - Jamming and Spoofing Capabilities
 - Provide real-time emulation of GNSS environments, including satellite positions, transmitted data, and timing
 - Signal Propagation & Errors Simulation
 - Propagation Effects: Multipath, Ionospheric and Tropospheric Models, and Terrain Modeling
- Error Introduction and Management:
 - Additive Pseudorange Ramps and Ephemeris Errors
 - Satellite Clock Error Modification and Navigation Message Errors

- Antenna Pattern Models.
 - Signal Dynamics
 - Max Relative Velocity: Up to 1,500,000 m/s
- Receiver Trajectory Simulation
 - Static, circular, car trajectories with integrated maps, and spacecraft orbits (LEO/GEO)
 - Import Capabilities: Allows for the import of arbitrary tracks/routes from formats like NMEA, CSV, or KML files.
 - Hardware-in-the-Loop (HIL)
 - Hardware Feature Description
 - Inputs
 - 10 MHz, 1 PPS, GPS Ant
 - Frequency Range 32 to 3225 MHz
 - Transmit Channels 4 minimum
 - Can simulate all constellations and frequency bands.
 - Emulate the following systems:
 - GPS Open: L1C/A, L1C, L1P, L2P, L2C, L5
 - GPS Encrypted: P(Y) Code, M-Code (AES, SDS, MNSA)
 - GLONASS: G1, G2
 - Equipment training is desired

Supply Chain Requirements:

The vendor shall include:

- A Certificate of Conformance (CoC) affirming that all supplied items meet the specifications outlined in this document.
- All components provided under this requirement shall be compliant with the Trade Agreements Act (TAA).
- The vendor shall certify that all end products and components originate from TAA-designated countries, in accordance with applicable federal regulations.

Notification of Commencement: The Contractor shall notify the TDN Lab Manager/POC identified above at the time of shipping.